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Table 7.2: Primary survey findings and triage categories – children

Criterion	Category 1 Immediate	Category 2 Emergency	Category 3 Urgent	Category 4 Semi-urgent	Category 5 Non-urgent
Airway	▪ Obstruction ▪ Partial obstruction (e.g. stridor, drooling) with severe respiratory distress ▪ Immediate risk of obstruction	▪ Partial obstruction (e.g. stridor, drooling) with moderate respiratory distress ▪ Imminent risk of obstruction	▪ Partial obstruction (stridor only with activity, no drooling) with minimal respiratory distress	No signs of obstruction	No signs of obstruction
Breathing	▪ Respiratory arrest ▪ Hypoventilation ▪ Signs of severe respiratory distress: – Severe use of accessory muscles – Severe intercostal and subcostal retraction – Acute cyanosis	▪ Signs of moderate respiratory distress: – Moderate use of accessory muscles – Moderate retraction – Tachypnoea – Pallor	▪ Signs of mild respiratory distress: – Mild use of accessory muscles – Mild retraction – Normal perfusion	No signs of respiratory distress	No signs of respiratory distress
Circulation	▪ Cardiac arrest or severe bradycardia (e.g. HR <60 bpm in an infant) ▪ Signs of severe shock: – Absent peripheral pulses – Pallor, cold, clammy, mottled – Severe tachycardia – Capillary refill >4 secs ▪ Uncontrolled bleeding	▪ Signs of shock: – Weak/thready peripheral pulses – Pallor, cool – Moderate tachycardia – Capillary refill 3–4 secs ▪ Signs of severe dehydration [>5 signs]* ▪ Severe blood loss	▪ Mild haemodynamic compromise: – Palpable peripheral pulses – Pallor, warm – Mild tachycardia – Capillary refill 2 secs ▪ Signs of moderate dehydration [3–5 signs]* ▪ Moderately severe blood loss	▪ No signs of haemodynamic compromise ▪ Less than 3 signs of dehydration* ▪ Mild haemorrhage	▪ No signs of haemodynamic compromise ▪ No signs of dehydration*

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Table 7.2: Primary survey findings and triage categories – children (continued)

Criterion	Category 1 Immediate	Category 2 Emergency	Category 3 Urgent	Category 4 Semi-urgent	Category 5 Non-urgent
Disability	▪ Unresponsive or responds to pain only ▪ Ongoing/prolonged seizure	▪ Decreased conscious state: GCS <13 ▪ Severe decrease in activity/change in behaviour: – Lethargic (quiet, disinterested in surroundings, goes back to sleep after being disturbed) – No eye contact – Decreased muscle tone ▪ Focal neurological signs ▪ Severe pain ▪ BGL <3 mmol/L	▪ Head injury with short LOC, now alert ▪ Moderate decrease in activity/change in behaviour: – Lethargic (quiet, tired looking) – Eye contact when disturbed ▪ Moderately severe pain	▪ Normal/no acute change to usual GCS ▪ Mild decrease in activity or change in behaviour: – Quiet but eye contact and interacts with parents ▪ Moderate pain	▪ Normal/no acute change to usual GCS ▪ No signs of lethargy or decrease in activity or change in behaviour: – Smiling, playing ▪ Mild or no pain
Exposure		Fever with signs of lethargy			
Other (secondary survey red flags)		▪ Major fracture ▪ Amputation (including partial) ▪ Suspected sepsis (physiologically unstable) ▪ Suspected testicular torsion ▪ Overdose or toxic ingestion	▪ Parietal, occipital or temporal haematoma following a fall in a child under 2 years ▪ Limb – altered sensation, acutely absent pulse ▪ Stable neonate ▪ Sepsis (stable) ▪ Child at risk of abuse/suspected non-accidental injury		

* Signs of dehydration: Decreased tissue turgor; decreased capillary refill time; sunken eyes; absent tears; dry mucous membranes, decreased urine output or decreased wet nappies, tachycardia, weak pulse, reduced activity and looking unwell

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Brain injury

Head injuries are common in children and most are not serious. Table 7.3, right, lists red flags for serious brain injury in children.

Pain

Pain determines urgency, but many children are too young or unwell to report their pain. Under these circumstances, a parent's assessment is usually the most efficient way to obtain an indication of the child's pain to determine clinical urgency. Keep in mind that a stoic child may have a significant injury but state they are not in pain. Cultural factors can also impact how a child expresses pain.

The impact of pain on the child may compromise your ability to assess or interpret signs of significant illness or injury. You should be cautious when assuming that pain and/or distress is the cause of vital sign derangements.

Exposure

The immune system is immature in neonates, infants and young children, which makes them more susceptible to infection and sepsis. However, they may not exhibit the classic signs of sepsis, such as fever and low blood pressure. Although fever should be considered a risk factor for serious illness in infants, hypothermia is a particularly concerning sign in this age group.

Rashes are a common feature of illness in infants and young children and are often worrying for parents. Rashes should be viewed at triage to identify non-blanching (petechial or purpuric) rashes. Also look for rashes suggesting an infectious disease, such as chicken pox or measles, which may require infection control actions.

Table 7.3: Red flags for serious brain injury in children

Age group	Feature
All	▪ GCS of 14 or less, or other signs of altered mental status ▪ Severe mechanism of injury ▪ Post-traumatic seizures ▪ Abnormal neurological examination
Children aged less than 2 years	▪ Palpable skull fracture* ▪ Occipital, parietal or temporal scalp haematoma† ▪ History of loss of consciousness for 5 secs or more ▪ Not acting normally, according to parent
Children aged 2 years and older	▪ Signs of base of skull fracture§ ▪ History of loss of consciousness ▪ History of vomiting# ▪ Severe headache

Source: Adapted from *Australia and New Zealand Guideline for Mild to Moderate Head Injuries in Children*[16]

* Palpable skull fracture: on palpation or possibly based on swelling or distortion of the scalp

† Non-frontal scalp haematoma: occipital, parietal or temporal

§ Signs of base of skull fracture: haemotympanum, 'raccoon eyes', cerebrospinal fluid (CSF) otorrhoea or CSF rhinorrhoea, Battle's sign

Isolated vomiting, without any other risk factors, is an uncommon presentation of clinically important traumatic brain injury. Vomiting, regardless of the number of episodes or persistence, in association with other risk factors, increases concern for clinically important traumatic brain injury.

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Assessment tools

The Paediatric Assessment Triangle (PAT) is an example of a paediatric-specific rapid assessment that can help identify children requiring urgent care.[17] It has three components: appearance, work of breathing and circulation to the skin.

Paediatric early warning scores (PEWS) are designed to detect signs of deterioration and allow for early intervention. Their value to triage decision-making has not been fully explored, but it is likely that low scores should not be considered reassuring.[18] However, they point to important parameters for assessment and thresholds for escalation.

The vital signs components are the basis for the track-and-trigger systems in place in most hospitals. High PEWS scores or a breach in track-and-trigger thresholds are indicative of serious illness and potential for rapid deterioration.[19]

Focused assessment

Once you have completed the primary survey and identified or excluded immediate and imminent life-threatening conditions, collect a brief past history and demographic details (see Table 7.4, right) and targeted history of the presenting problem (see Table 7.5). This will help identify risk factors for serious illness or injury and the potential for deterioration.

Past history and other demographic factors should also be evaluated for their potential impact on the child's condition and risk of deterioration.

Table 7.4: Past history and demographics

Topic	Notes
Age, prematurity and low birth weight	Neonates, infants and other younger children are particularly vulnerable to serious illness, including sepsis and respiratory illness, and those born prematurely are at additional risk[6]
Pre-existing illnesses, including congenital diseases	These increase the likelihood of contracting specific illnesses and the risk of more severe disease and poor outcomes (e.g. cardiac disease increases the risk of cardiorespiratory dysfunction associated with minor respiratory illnesses)[2]
Allergies	Allergies are extremely common and allergen exposure may account for the presentation
Vaccination status	Unvaccinated or partially vaccinated children are at increased risk of vaccine-preventable diseases – many of which are linked to serious illness
Socioeconomic factors	Children from socially disadvantaged and marginalised communities are at greater risk of serious illness and injury[20]

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Table 7.5: Targeted history – the presenting problem

Topic	Notes
Oral intake	Decreases should trigger concern regarding hydration, particularly when coupled with significant losses (e.g. vomiting and diarrhoea); decreased feeding is also a sign of serious illness and an indirect indication of respiratory distress in infants[11,12]
Behaviour and activity levels	Acute or recent changes in behaviour (e.g. irritability) and reduction in activity may indicate significant illness and injury[2,20]
Mechanism of injury	Prediction of potential injuries is important as children may not exhibit obvious signs of injury and are likely to maintain relatively normal vital signs until substantial blood volume is lost
Medication or toxin ingestion	Assume that a young child has ingested the largest dose possible; if more than one child is involved, treat each child as if they ingested the total dose
Sick contacts	May identify children at increased risk of infectious illness, placing them at risk of serious illness, and other children at risk of hospital-acquired infection[2]
Risk assessment	Suicide and self-harm risk should be assessed in children and young people presenting with behavioural disturbance and mental health concerns

Other considerations

Unaccompanied minors

Children may present to the ED unaccompanied for various reasons, including self-referral and ambulance transfer following trauma. This may limit the information available, increase the child's distress and raise concerns about consent and the safety of the child while waiting (for example, undetected deterioration and leaving before being seen). Consider these factors and local processes when assessing urgency and suitable care environment for the child.

Family violence and children protection issues

These issues are common and may be identified during an ED presentation. Risk factors for family violence and/or child abuse include:

- Infants: injuries in a non-ambulant child, specific patterns consistent with inflicted injuries, unsettled infants, signs of poor attachment and avoidant gaze
- Children and adolescents: recurrent unexplained symptoms, regressive behaviours, poor social skills, erratic school attendance, mental health problems, substance use
- Caregivers: inconsistent history of circumstances, significant injuries attributed to another child, delayed presentation, previous contact with protective services, parental distress inconsistent with presentation, one parent dominating the interaction.

Health professionals are required by law to report to child protection services signs that a child or young person is at risk. If suspicion is raised at triage, follow local policies to prompt further enquiry and reporting.

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Red flags

Features of possible serious illness and risk of rapid deterioration in children include:

- Signs of respiratory distress
- Acute pallor
- Decreased activity or drowsiness
- Floppiness
- Inconsolable distress
- Decreased fluid intake (for example, less than half the usual volume in the past 24 hours) and output (such as less than four wet nappies in the past 24 hours)
- Looking unwell
- Parental concern
- Re-presentation
- Age less than 3 months and fever or temperature less than 36 °C.[5-7]

See Table 7.1, right, for selected low-frequency but high-risk presentations in children. Any of these features should prompt further, careful assessment and consideration of higher urgency.

Table 7.1: Selected low-frequency, high-risk paediatric presentations

Feature	Comment
Pale, floppy episodes in infants	Linked to serious illnesses, such as anaphylaxis and intussusception
Abdominal pain of sudden onset	Indicates risk of conditions needing urgent surgical treatment, including torsion of the testes or ovary
Tachycardia, especially if afebrile	Linked to sepsis
Bilious (green) vomiting	Indicates bowel obstruction
Exercise-induced collapse	Likely cardiac cause
Sudden onset of neurological symptoms in a previously well child	Linked to stroke
Magnets and button batteries	High-risk foreign bodies requiring urgent removal
Non-blanching rash	Associated with meningococcal disease
Severe mechanism of injury	e.g. fall >2 m, MVC* >60 km/h, ejection/rollover, prolonged extrication (>30 minutes), death of same-car occupant, pedestrian, MBC**/cyclist >30 km/h
Expressing intent to harm self or others	Suggests a triage category 2. See **Chapter 5: Psychological distress and behavioural disturbance**
Delayed presentation **History inconsistent with the presentation**	May suggest the child is at risk of harm

*Motor vehicle collision **Motorbike collision